

CLASS 1

AI coding for VFX artists

Get started with Claude Code, Codex and Antigravity. No programming background needed.

John Huikku

WHY THIS IS WORTH YOUR TIME

You already know
the core skill: **giving notes.**

You brief, review and approve work every day. Working with an AI assistant is the same loop. It just moves faster.

THE TOOLS

What these tools actually do



It reads

Your files and folders, clip specs, error messages and screenshots.



It acts

Renames, converts, writes scripts, runs ffmpeg, and drives apps through MCP.



You approve

It asks before it touches anything. You review the result like a dailies note.

Think of it as a fast junior artist who never gets tired **and still needs supervising.**

THE TOOLS

Three tools, three ways in

Use the one your subscription already includes. **Everyone starts in the desktop app.**

Tool	Comes with	Desktop app	Editor	Terminal (CLI)
Claude Code	Claude Pro, Max, Team	Claude app, Code tab	VS Code extension	Claude Code CLI
Codex	Any ChatGPT plan	ChatGPT app, Codex mode	VS Code extension	Codex CLI
Antigravity	Free, or Google AI Pro	Antigravity app	Antigravity IDE	Antigravity CLI
Best for		Everyone. Start here.	Keeping your own scripts	Automation, power users

THE TOOLS

You'll hear other names too

Chat apps

ChatGPT · Claude · Gemini

Answers in a chat window. You copy and paste the results yourself.

Agents

Claude Code · Codex · Antigravity

Also: GitHub Copilot, OpenCode.
They work directly in your files and apps.

Editors

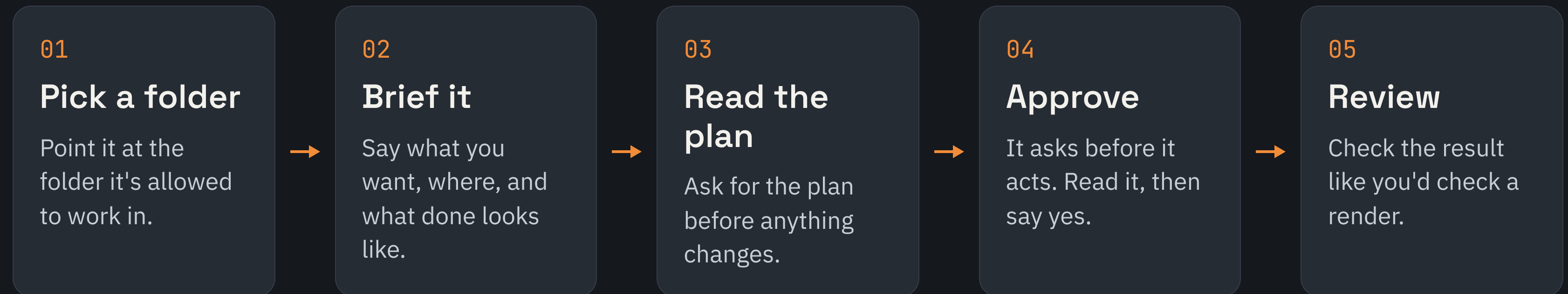
VS Code · Cursor · Windsurf

For people writing and keeping code. Cursor and Windsurf have AI built in.

The jump that matters: **from a chat window to an assistant working in your files.**

THE TOOLS

How a session works



Not right? Give a note and go around again. Same as dailies.

GOOD HABITS

Five habits that keep you safe

- 1 Work on copies**
Point it at a copy or a test folder until you trust the task.
- 2 Dry run first**
Ask it to list what it would change before it moves, renames or deletes anything.
- 3 Read before you approve**
Keep approvals on. Don't switch on auto-approve until you're comfortable.
- 4 Watch what you paste**
No passwords, API keys or confidential project details in prompts or file names.
- 5 Only trusted add-ons**
MCPs and skills run code on your machine. Install them only from sources you trust.

GOOD HABITS

Brief it like you'd note a shot

Vague

`convert these videos`

Which videos? To what? Where do they go?
Can it overwrite the originals?

Clear

`In the review_copies folder, convert every .mov to
H.264 .mp4 at 1920x1080. Keep the original file names.
Show me the plan first, and don't delete anything.`

Say where

Say what done looks like

Give an example

Ask for the plan first

GOOD HABITS

Stuck? Ask it to teach you

Phrases that unlock it

Explain this like I'm an artist, not a coder.

Ask me questions before you start.

What does this error mean? What do I do?

Teach me as you go.

What should I learn next to do this myself?

Make it prove it

It can say "done" when it isn't. Ask it to show the file, play the output or open the page.

When it goes sideways

- Start a fresh session
- Give it more context
- Break the task into smaller steps
- Try another model or tool

WHAT YOU CAN DO TODAY

Organize: hand off the tedious part

Inventory

List every clip in this folder with its duration, resolution and frame rate. Save it as a spreadsheet.

Rename

Rename these renders to SHOW_SHOT_v###.exr. Show me a dry-run table of old and new names first.

Sort

Sort my downloads into plates, references, renders and other. Move nothing until I approve the list.

Anything that moves or renames files: **dry run first, every time.**

WHAT YOU CAN DO TODAY

Convert: check the output, then give a note

1 • The ask

Convert this phone clip to H.264 for review.



2 • The result

Washed out and too bright. The phone shot HDR, and a plain conversion ignores that.



3 • The note

This looks washed out. It's HDR phone footage. Tone-map it to normal video.

Other quick wins

Image sequence to MP4

Proxies and review copies

Contact sheets

GIFs for chat

Pull audio from video

WHAT YOU CAN DO TODAY

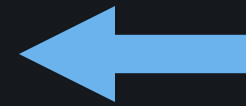
The movie never went to the cloud



Your machine: the hands

The footage stays here. The script runs here, and ffmpeg does the actual conversion locally.

Your brief, file names, command output



A script or command to run



The AI model: the brain

Plans the job and writes the Python or shell script. It never receives the movie file.

Everything that came back for the HDR clip (simplified):

```
ffmpeg -i clip.mov -vf "zscale=t=linear,tonemap=hable,zscale=t=bt709" -c:v libx264 clip_review.mp4
```

 **What does go up:** your prompt, file names, error messages, text files it opens, and any frame or screenshot it looks at.

WHAT YOU CAN DO TODAY

Any DCC, today: ask for a script

No special setup. Ask for it, paste it into the app's script editor, run it.

Nuke

Write a Nuke Python script that sets every selected Read node to the project frame range.

Maya

Write a Maya script that renames the selected objects with a prefix and padded numbers.

Houdini

Write a Houdini shelf tool that adds a camera and a basic three-light rig.

After Effects

Write an After Effects script that lists every comp's name and duration in a text file.

Next level: Nuke, Maya and Houdini all ship PySide. Ask for a tool panel with buttons.

WHAT YOU CAN DO TODAY

Real example: a copper shader from one prompt

1 • The ask

Write a Maya Python script that builds a copper material for Arnold from scratch and assigns it to the selection.

2 • The script

Python that builds the whole shading network: color, metalness, roughness, all wired up.

3 • Run it in Maya

Pasted into the Script Editor. The network built itself, and it looked just like copper.



Made by a script, rendered headless (here in Blender). The same idea runs in Maya with mayapy.

WHAT YOU CAN DO TODAY

One script, two ways to run it

In the app

- Paste it into the script editor and run
- One scene, and you watch it happen
- Best for trying and tweaking



the same .py file

Headless (no window)

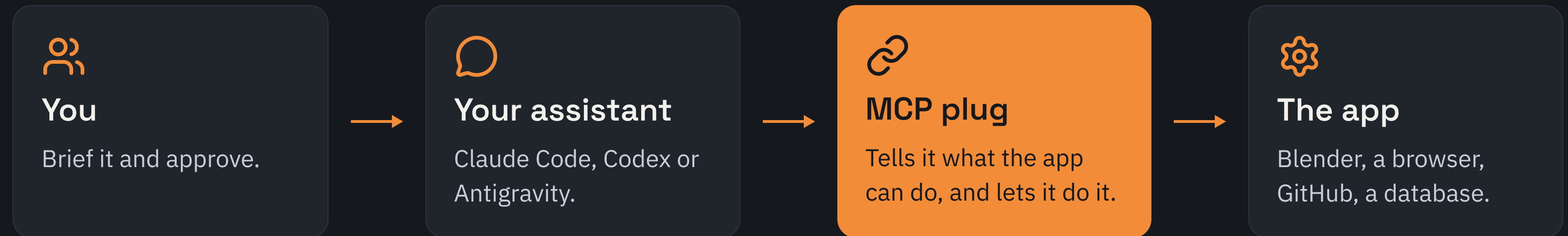
- Run from the command line; the app never opens
- Hundreds of files, or another machine overnight
- Best once the script works

```
# Blender: build a scene from scratch and save it
blender -b --python build_copper_scene.py -- out/copper.blend
# Blender: run a script on an existing file
blender -b shot_010.blend --python fix_materials.py
```

Same idea elsewhere: mayapy (Maya) · hython (Houdini) · nuke -t · c4dpy (Cinema 4D) · 3dsmaxbatch

WHAT YOU CAN DO TODAY

MCP: a plug that lets it drive your apps



Without MCP

It writes a script. You paste and run it.

With MCP

It works in the app directly: builds the scene, sets the lights, renders.

 An MCP runs code on your machine. Install only ones from sources you trust.

WHAT YOU CAN DO TODAY

MCP for Blender in action

The prompt

In Blender, set up a three-point light rig and a turntable camera around the selected object.

With MCP, the assistant works inside Blender itself: it adds the key, fill and rim lights one by one, then sets up the turntable.



Setup: setup prompt 5 in the class folder · github.com/ahujasid/mcp-for-blender

KEEP YOUR WORK

GitHub basics, in VFX terms

Term	What it is	VFX version
Repo	A project folder that remembers every change	Show folder with every version
Commit	A save point, with a note on what changed	Versioning up v003 to v004
Push	Send your commits up to GitHub	Publishing to the server
Pull	Get the latest changes from GitHub	Syncing to latest
Clone	Download a repo, full history included	Copying down the whole project
Branch	A side version for trying an idea	A wip you can throw away
Merge	Bring a branch back into the main version	Promoting the wip to hero
Pull request	"Please review this and merge it"	Submitting for dailies
Fork	Your own copy of someone else's repo	Starting from a studio template
README	The repo's front page: what it is, how to use it	The show bible
.gitignore	A list of files git must never save	"Never publish caches or renders"

KEEP YOUR WORK

Private by default

Never put in a repo

- Passwords, API keys, tokens
- Client or project data
- People's names and emails
- Internal paths, sheet IDs, Drive links
- Footage, renders, EXRs (media stays on Drive)

Deleting isn't erasing

History keeps old versions. If a secret was ever pushed, treat it as leaked and replace it.

Public only after a cleanup

Start every repo private. Go public only when it's fully sanitized, on purpose.

```
Commit this with a short note and push it to my
private repo.
```

WHERE IT GOES NEXT

Which stack for what

Building...	Use
A quick utility: rename, convert, organize	Python + uv + ffmpeg
A utility with a simple UI	Python + Gradio
A tool inside a DCC	Python + PySide6, or an MCP
A desktop app for your team	Tauri + the web UI stack, or PySide6 + PyInstaller
A web app	React · Vite · Tailwind · shadcn, hosted on GitHub Pages, Netlify or Vercel
...that uses an API key	+ a serverless function or a Railway backend
...that needs logins or a database	+ Supabase
...that stores lots of media	+ Backblaze B2 or Cloudflare R2

WHERE IT GOES NEXT

Web apps: why keys live on the backend

Frontend

GitHub Pages · Netlify · Vercel. Anyone can look inside.



Backend

Railway or a serverless function. Holds the API keys.



Supabase

Data, logins

B2 · R2

Media

APIs

fal.ai, weather

Frontend = the dining room. Everyone walks through it.

Backend = the kitchen. Staff only.

API key = the restaurant's credit card with its suppliers.

Full stack = the whole restaurant: frontend, backend and database together.

Frontend asks.
Backend spends.

Set a spending limit on every API account.

WHERE IT GOES NEXT

Start tonight: your class folder

01

Get it

Clone it, or use Download ZIP. Keep it on your own drive, not in Drive or OneDrive.

02

Open it

Open the folder in Claude Code, Codex or Antigravity. The rules and skills load by themselves.

03

Say "go"

It becomes your tutor: asks your level, makes practice files, picks your first project.

Get it at github.com/huikku/ai-coding-for-vfx-class

Updates: "Pull the latest class updates." That's git pull, and your notes and practice files are never touched.

Your real work goes in its own project folder, set up with the starter pack.

WHERE IT GOES NEXT

The starter pack: studio rules for your assistant

It forgets between sessions. These files remind it, every time. The class folder already has them; copy the pack into your own projects.

AGENTS.md

Studio rules: naming, folders, codecs, "dry-run first." All three tools read it.

Skills

Reusable how-tos: safe file moves, HDR-safe conversions, contact sheets, DCC scripts.

.gitignore

Keeps media, caches and keys out of GitHub automatically.

Setup prompts

Paste one in and it installs ffmpeg, GitHub, MCP for Blender or Tauri for you.

WHERE IT GOES NEXT

Your first week

Pick one starter project

1. List the specs of every clip in a folder
2. Rename a render folder to convention (dry run first)
3. Turn an image sequence into an MP4
4. Make a contact sheet of a shot folder
5. A script for your DCC's most boring task
6. A Gradio app that runs a fal.ai image tool
7. Put it all in a private GitHub repo

Stuck? Ask in this order

1 · Your AI assistant

"Explain this like I'm an artist."

2 · The class folder

Open it in your assistant: tutor mode

3 · The docs

PDFs and prompts in the class folder

4 · Ask John

Once you've tried 1 to 3. Share your wins too!

You don't need to know how.
You need to know what to ask.

Open the class folder · say "go" · pick a first project · stuck? ask your AI first

Questions?

John Huikku